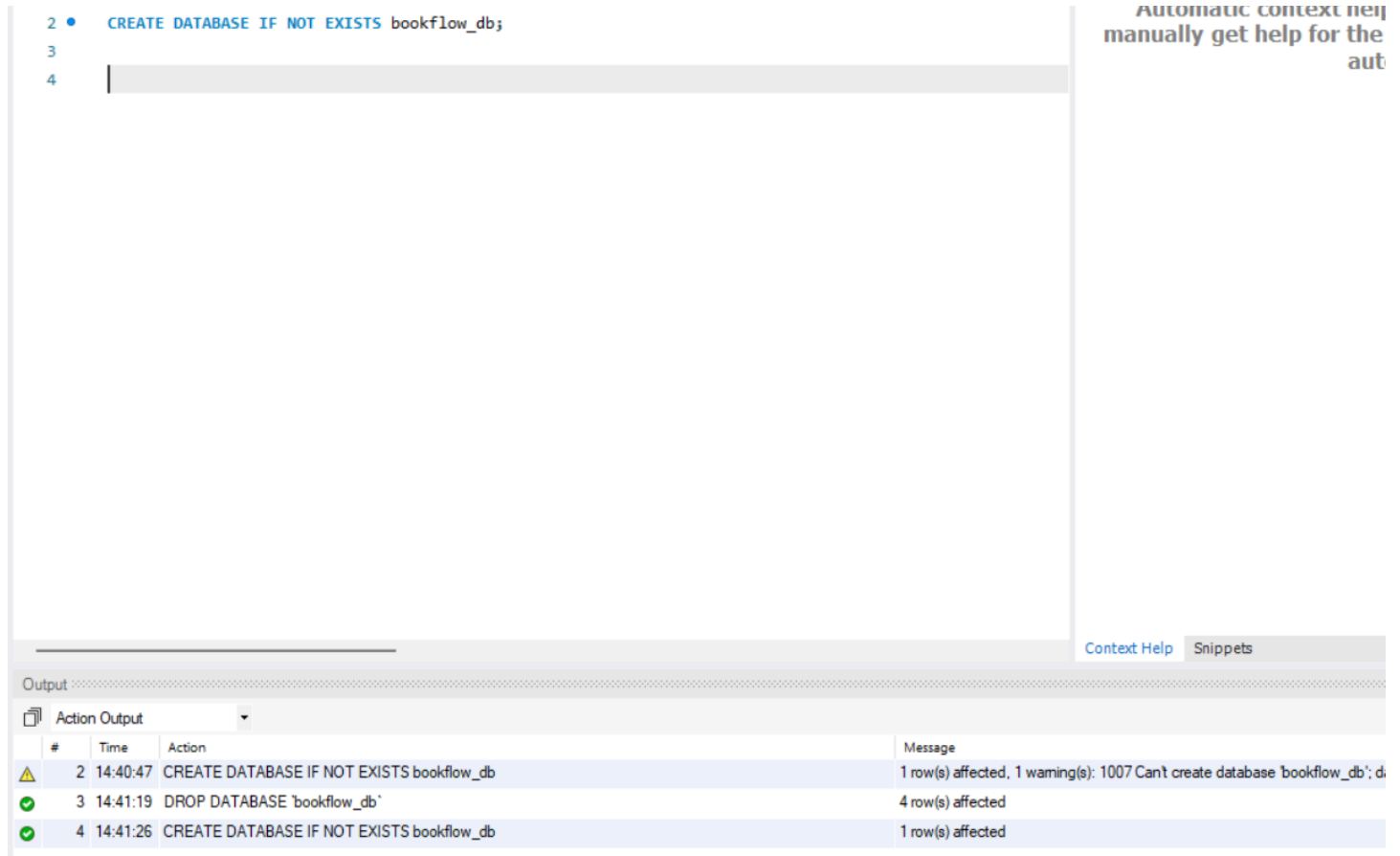


2520030178_Aasritha

WEEK-1 PRACTICAL DATABASE SYSTEM ENGINEERING

Query : 1 CREATE DATABASE IF NOT EXISTS bookflow_db;

output:



The screenshot shows a database management interface. At the top, a query window contains the SQL command: `CREATE DATABASE IF NOT EXISTS bookflow_db;`. Below the query window, an output log displays the execution results. The log is titled "Output" and shows a table with columns: #, Time, Action, and Message. The table contains three rows of data, indicating the execution of the query, a warning, and the successful creation of the database.

#	Time	Action	Message
2	14:40:47	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bookflow_db'; d
3	14:41:19	DROP DATABASE 'bookflow_db'	4 row(s) affected
4	14:41:26	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected

Query 2: USE bookflow_db;

output:

2 • CREATE DATABASE IF NOT EXISTS bookflow_db;
3 • USE bookflow_db;
4
5

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Context HelpSnippets

Output :
Action Output

#	Time	Action	Message
3	14:41:19	DROP DATABASE 'bookflow_db'	4 row(s) affected
4	14:41:26	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected
5	14:48:14	USE bookflow_db	0 row(s) affected

Query 3: CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL, isbn VARCHAR(13) NOT NULL UNIQUE, published_year INT, CONSTRAINT chk_published_year CHECK (published_year < 2027));

output:

```
2 • CREATE DATABASE IF NOT EXISTS bookflow_db;  
3 • USE bookflow_db;  
4 • CREATE TABLE books (  
5     book_id  
6     INT AUTO_INCREMENT PRIMARY KEY,  
7     title  
8     VARCHAR(255) NOT NULL,  
9     isbn  
10    VARCHAR(13) NOT NULL UNIQUE,  
11    published_year INT,  
12    CONSTRAINT chk_published_year CHECK (published_year < 2027)  
13 );  
14  
15
```

[Context Help](#) [Snipp](#)

Output

Action Output

#	Time	Action	Message
✓ 4	14:41:26	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected
✓ 5	14:48:14	USE bookflow_db	0 row(s) affected
✓ 6	14:49:23	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) ...	0 row(s) affected

Query 4: Create the members table CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, email VARCHAR(150) NOT NULL UNIQUE);

output:

```

2 • CREATE DATABASE IF NOT EXISTS bookflow_db;
3 • USE bookflow_db;
4 • CREATE TABLE books (
5     book_id
6     INT AUTO_INCREMENT PRIMARY KEY,
7     title
8     VARCHAR(255) NOT NULL,
9     isbn
10    VARCHAR(13) NOT NULL UNIQUE,
11    published_year INT,
12    CONSTRAINT chk_published_year CHECK (published_year < 2027)
13 );
14 • CREATE TABLE members (
15     member_id INT AUTO_INCREMENT PRIMARY KEY,
16     full_name VARCHAR(100) NOT NULL,
17     email
18     VARCHAR(150) NOT NULL UNIQUE
19 );
20

```

[Context Help](#)

Output

Action Output

#	Time	Action	Message
✓ 5	14:48:14	USE bookflow_db	0 row(s) affected
✓ 6	14:49:23	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) ...	0 row(s) affected
✓ 7	14:50:43	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(100)...	0 row(s) affected

Query 5 (verification): Inspect the table structures DESCRIBE members;
output:

```

16 full_name VARCHAR(100) NOT NULL,
17 email
18 VARCHAR(150) NOT NULL UNIQUE
19 );
20 • DESCRIBE members;

```

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	auto_increment
full_name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	

Result 1 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message
6	14:49:23	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255)...	0 row(s) affected
7	14:50:43	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(10...	0 row(s) affected
8	14:51:40	DESCRIBE members	3 row(s) returned

Query 6: Insert 3 books INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Clean Code', '9780132350884', 2008), ('Atomic Habits', '9780735211292', 2018);

output:

The screenshot displays a SQL IDE interface with a script editor and an output pane. The script editor contains the following SQL commands:

```
3 • USE bookflow_db;
4 • CREATE TABLE books (
5     book_id
6     INT AUTO_INCREMENT PRIMARY KEY,
7     title
8     VARCHAR(255) NOT NULL,
9     isbn
10    VARCHAR(13) NOT NULL UNIQUE,
11    published_year INT,
12    CONSTRAINT chk_published_year CHECK (published_year < 2027)
13 );
14 • CREATE TABLE members (
15     member_id INT AUTO_INCREMENT PRIMARY KEY,
16     full_name VARCHAR(100) NOT NULL,
17     email
18     VARCHAR(150) NOT NULL UNIQUE
19 );
20 • DESCRIBE members;
21 • INSERT INTO books (title, isbn, published_year) VALUES
22 ('The Alchemist', '9780061122415', 1988),
23 ('Clean Code', '9780132350884', 2008),
24 ('Atomic Habits', '9780735211292', 2018);
25
```

The output pane shows the results of the executed queries:

#	Time	Action	Message
7	14:50:43	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(10...	0 row(s) affected
8	14:51:40	DESCRIBE members	3 row(s) returned
9	14:52:43	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Clean ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Query 7: Insert 3 members INSERT INTO members (full_name, email)
VALUES ('Anil Kumar', 'anil.kumar@example.com'), ('Priya Sharma',
'priya.sharma@example.com'), ('Ravi Verma', 'ravi.verma@example.com');
output:

Limit to 1000 rows

```

6  INT AUTO_INCREMENT PRIMARY KEY,
7  title
8  VARCHAR(255) NOT NULL,
9  isbn
10 VARCHAR(13) NOT NULL UNIQUE,
11 published_year INT,
12 CONSTRAINT chk_published_year CHECK (published_year < 2027)
13 );
14 CREATE TABLE members (
15 member_id INT AUTO_INCREMENT PRIMARY KEY,
16 full_name VARCHAR(100) NOT NULL,
17 email
18 VARCHAR(150) NOT NULL UNIQUE
19 );
20 DESCRIBE members;
21 INSERT INTO books (title, isbn, published_year) VALUES
22 ('The Alchemist', '9780061122415', 1988),
23 ('Clean Code', '9780132350884', 2008),
24 ('Atomic Habits', '9780735211292', 2018);
25 INSERT INTO members (full_name, email) VALUES
26 ('Anil Kumar', 'anil.kumar@example.com'),
27 ('Priya Sharma', 'priya.sharma@example.com'),
28 ('Ravi Verma', 'ravi.verma@example.com');

```

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Context Help Snippets

Output

Action Output

#	Time	Action	Message
8	14:51:40	DESCRIBE members	3 row(s) returned
9	14:52:43	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Clean ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
10	14:53:38	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com'), ('Priya Sharma...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Test 1: Duplicate ISBN (UNIQUE violation) INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000);

output:

```

20 DESCRIBE members;
21 INSERT INTO books (title, isbn, published_year) VALUES
22 ('The Alchemist', '9780061122415', 1988),
23 ('Clean Code', '9780132350884', 2008),
24 ('Atomic Habits', '9780735211292', 2018);
25 INSERT INTO members (full_name, email) VALUES
26 ('Anil Kumar', 'anil.kumar@example.com'),
27 ('Priya Sharma', 'priya.sharma@example.com'),
28 ('Ravi Verma', 'ravi.verma@example.com');
29
30 INSERT INTO books (title, isbn, published_year)
31 VALUES ('Fake Copy', '9780061122415', 2000);

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message
9	14:52:43	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Clean ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
10	14:53:38	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com'), ('Priya Sharma...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
11	14:54:43	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'

Test 2: NULL title (NOT NULL violation) INSERT INTO books (title, isbn, published_year) VALUES (NULL, '99999999999999', 2010);

output:

exp1* customer_transactions SQL File 4* SQL File 5* SQL File 6* bookflow_db - Schema SQLAdditions

Limit to 1000 rows

```

12  CONSTRAINT chk_published_year CHECK (published_year < 2027)
13  );
14  CREATE TABLE members (
15  member_id INT AUTO_INCREMENT PRIMARY KEY,
16  full_name VARCHAR(100) NOT NULL,
17  email
18  VARCHAR(150) NOT NULL UNIQUE
19  );
20  DESCRIBE members;
21  INSERT INTO books (title, isbn, published_year) VALUES
22  ('The Alchemist', '9780061122415', 1988),
23  ('Clean Code', '9780132350884', 2008),
24  ('Atomic Habits', '9780735211292', 2018);
25  INSERT INTO members (full_name, email) VALUES
26  ('Anil Kumar', 'anil.kumar@example.com'),
27  ('Priya Sharma', 'priya.sharma@example.com'),
28  ('Ravi Verma', 'ravi.verma@example.com');
29
30  INSERT INTO books (title, isbn, published_year)
31  VALUES ('Fake Copy', '9780061122415', 2000);
32
33  INSERT INTO books (title, isbn, published_year)
34  VALUES (NULL, '9999999999999', 2010);

```

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Context Help Snippets

Output

Action Output

#	Time	Action	Message
✓ 10	14:53:38	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com'), ('Priya Sharma...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✗ 11	14:54:43	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'boo
✗ 12	14:55:52	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999', 2010)	Error Code: 1048. Column 'title' cannot be null

Test 3: Future publication year (CHECK violation) INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '88888888888888', 2030);

output:

The screenshot shows a SQL IDE interface with a script editor and an output window. The script editor contains the following SQL code:

```
14 CREATE TABLE members (  
15     member_id INT AUTO_INCREMENT PRIMARY KEY,  
16     full_name VARCHAR(100) NOT NULL,  
17     email  
18     VARCHAR(150) NOT NULL UNIQUE  
19 );  
20 DESCRIBE members;  
21 INSERT INTO books (title, isbn, published_year) VALUES  
22 ('The Alchemist', '9780061122415', 1988),  
23 ('Clean Code', '9780132350884', 2008),  
24 ('Atomic Habits', '9780735211292', 2018);  
25 INSERT INTO members (full_name, email) VALUES  
26 ('Anil Kumar', 'anil.kumar@example.com'),  
27 ('Priya Sharma', 'priya.sharma@example.com'),  
28 ('Ravi Verma', 'ravi.verma@example.com');  
29  
30 INSERT INTO books (title, isbn, published_year)  
31 VALUES ('Fake Copy', '9780061122415', 2000);  
32  
33 INSERT INTO books (title, isbn, published_year)  
34 VALUES (NULL, '9999999999999', 2010);  
35 INSERT INTO books (title, isbn, published_year)  
36 VALUES ('Time Traveler', '8888888888888', 2030);
```

The output window shows the following errors:

#	Time	Action	Message
11	14:54:43	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'
12	14:55:52	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999', 2010)	Error Code: 1048. Column 'title' cannot be null
13	14:56:34	INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '8888888888888', 2030)	Error Code: 3819. Check constraint 'chk_published_year' is violated.

Test 4: Duplicate email (UNIQUE violation) INSERT INTO members (full_name, email) VALUES ('Anil Clone', 'anil.kumar@example.com');

output:

exp1*customer_transactionsSQL File 4*SQL File 5*SQL File 6*bookflow_db - Schema

Limit to 1000 rows

SQLAdditions

16full_name VARCHAR(100) NOT NULL,
17email
18VARCHAR(150) NOT NULL UNIQUE
19);
20• DESCRIBE members;
21• INSERT INTO books (title, isbn, published_year) VALUES
22('The Alchemist', '9780061122415', 1988),
23('Clean Code', '9780132350884', 2008),
24('Atomic Habits', '9780735211292', 2018);
25• INSERT INTO members (full_name, email) VALUES
26('Anil Kumar', 'anil.kumar@example.com'),
27('Priya Sharma', 'priya.sharma@example.com'),
28('Ravi Verma', 'ravi.verma@example.com');
29
30• INSERT INTO books (title, isbn, published_year)
31VALUES ('Fake Copy', '9780061122415', 2000);
32
33• INSERT INTO books (title, isbn, published_year)
34VALUES (NULL, '9999999999999', 2010);
35• INSERT INTO books (title, isbn, published_year)
36VALUES ('Time Traveler', '8888888888888', 2030);
37• INSERT INTO members (full_name, email)
38VALUES ('Anil Clone', 'anil.kumar@example.com');

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Context HelpSnippets

Output

Action Output

#	Time	Action	Message
✖12	14:55:52	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999', 2010)	Error Code: 1048. Column 'title' cannot be null
✖13	14:56:34	INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '8888888888888', 2030)	Error Code: 3819. Check constraint 'chk_published_year' is violated.
✖14	14:57:06	INSERT INTO members (full_name, email) VALUES ('Anil Clone', 'anil.kumar@example.com')	Error Code: 1062. Duplicate entry 'anil.kumar@example.com' for key 'members.email'